Fiscal Year:	FY 2006	**Task Last Updated:**	FY 10/02/2005
PI Name:	Urbach, Jeffrey S Ph.D.		
Project Title:	Non-equilibrium Dynamics and Statistical Mechanics of Excited Granular Media in the Absence of Gravitational Barriers		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Granular flows		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20057-1227	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	12/01/2003	**End Date:**	09/30/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (11/06)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Egolf, David (Georgetown University)		
Grant/Contract No.:	NNC04GA63G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In reduced gravity or microgravity environments, regimes without enduring contacts between particles are common. The gravitational barriers present on Earth tend to mask the effects that will dominate in a granular gas. We are undertaking a series of matched experiments and computer simulations on vibrated granular media in a geometry chosen to minimize the effects of gravity consisting of a collection of identical spherical particles confined between two parallel vibrating plates. This system is well suited to investigate many of the complex non-equilibrium effects observed in granular gases. Investigate the statistical mechanics of vibrated, unsheared monodisperse granular layers. Statistical characterization essential for developing continuum models of granular flows. We will measure the energy flow from the driving to the kinetic energy of the particles, diffusive transport of particles, and the nature of observed phase transitions. We will also construct a granular rheometer that incorporates an oscillating disk into the center of the vibrating container to measure the stress tensor for the granular fluid and measure changes in local average and fluctuating quantities induced by the imposed shear. This will allow stringent tests of the assumptions and predictions of continuum equations for granular flow, particularly because we will be able to control the shear rate and the kinetic energy of the particles independently. We will also use rheometery of dense shear flow to investigate the utility of some newly developed, potentially powerful and far reaching ideas from non-equilibrium statistical mechanics. We will investigate a newly discovered segregation mechanism in mixtures of particles of different sizes, similar to the depletion force observed in colloidal suspensions. This interaction, which is masked by gravitational effects in most terrestrial granular flows, could have significant implications for flows in microgravity. The closely matched experiments and computer simulations are designed to have a broad impact on our understanding of granular flow. By carefully investigating a well controlled model system, we will be able to develop and validate kinetic and hydrodynamic descriptions of granular media that incorporate the complex correlations observed in granular flow, characterize new mechanisms for interactions and segregation in granular mixtures, and test new fundamental concepts in non-equilibrium statistical mechanics that may have wide applicability for systems driven far from equilibrium.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We have measured a complex phase diagram that depends sensitively on the spacing between the horizontal plates that confine the granular layer. These behavior is observed in both the experiment and in the molecular dynamics simulations. A very similar phase diagram is observed in confined hard-sphere colloidal suspensions in equilibrium, but with some important differences that arise from the dissipation caused by inter-particle collisions. The significance of these results is that they indicate that the driving mechanism behind an equilibrium phase transition may still operate far from equilibrium, but that models of transitions in granular flow must take into account the lack of equipartition in the kinetic energy of the particles. We have completed a detailed study of the response of the granular layer to a steady applied shear. In both the experiments and the computer simulations, we have varied the density of spheres, the vibration amplitude, the applied shear rate, and the inelasticity of the spheres. Our results show that the confining plates have a strong influence on the observed shear profile, but that the behavior observed can be modeled with a hydrodynamic description. We are currently using these results to test the applicability of transport coefficients derived from kinetic theory.
Bibliography Type:	Description: (Last Updated: 12/21/2007)
Articles in Peer-reviewed Journals	Olafsen JS, Urbach JS. "Two-dimensional melting far from equilibrium in a granular monolayer." Phys Rev Lett. 2005 Aug 26;95(9):098002. PMID: 16197252 , Aug-2005
Articles in Peer-reviewed Journals	Melby P, Vega Reyes F, Prevost A, Robertson R, Kumar P, Egolf DA, Urbach JS. "The dynamics of thin vibrated granular layers." J. Phys. Cond. Mat. 2005 June 22;17(24): S2689-S2704 , Jun-2005
Articles in Peer-reviewed Journals	Prevost A, Melby P, Egolf DA, Urbach JS. "Nonequilibrium two-phase coexistence in a confined granular layer." Phys Rev E Stat Nonlin Soft Matter Phys. 2004 Nov;70(5 Pt 1):050301. PMID: 15600577 , Nov-2004